(a)



(b)



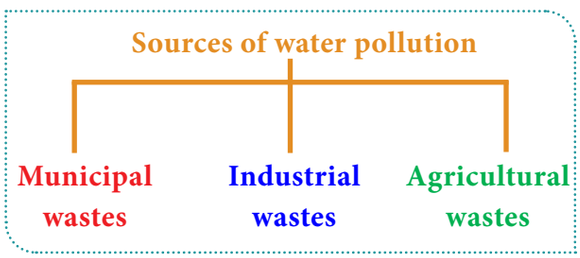
(c)



Fig 13.2 Sources of water pollution

Leaks and Spills: This occurs mostly due to ship collision, off shore oil rigs, oil leakages and discharges into sea **(Fig. 13.2 c).**

Sources of water pollution can also be classified in three ways. They are municipal wastes, industrial wastes, and agricultural wastes.



1. Municipal waste water is from homes and commercial establishments.
2. Industrial discharge (effluents) may contain varieties of compounds such as heavy metals (cadmium, chromium, lead), and organic / inorganic chemicals containing waste water, sometimes in toxic concentrations. These discharges can affect temperatures of the water bodies as well as dissolved oxygen level.
3. Agricultural wastes include fertiliser and pesticide runoff from agricultural fields, food processing waste, tree and saw dust from logging operations and bacteria from sewage or livestock operations.

Water pollutants reach water bodies like rivers, streams and the marine system by precipitation, run-off and the groundwater by seepage or percolation.

13.3.4 Effect of Water pollution on Ecosystems

1. **Destruction of ecosystems:** Ecosystems, especially aquatic systems, can be severely affected or destroyed by water pollution. Water pollutants affect existing niches and habitats and the survival of organisms. Soil fertility is affected and the system becomes uninhabitable.
2. **Disruption of food-chains:** Water pollution disrupts the natural food chains as well as food webs. Pollutants such as lead and cadmium are taken up by primary consumers where they can be lethal or get stored. Later, when these animals are consumed by secondary consumers, the food chain can get disrupted at any trophic level or result in enhanced concentration of these pollutants (biomagnification). Hot water from industries when released into the water bodies affects aquatic density and diversity.

13.3.5. Effect of Water pollution on Organisms

1. Water pollution can be lethal to aquatic organisms and others that depend on these water bodies.

Accidental oil spills from tanker ships can cause substantial environmental damage. Oil spreads on the water surface, prevents the entry of light and oxygen into the water. This increases BOD and COD, resulting in mass death of organisms

XII Std Zoology Chapter 13 EM.indd 224 23/12/2022 16:53:52